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Ph.D., Economics, Yale University, 2027 (expected)
M.Phil., Economics, Yale University, 2024
M.A., Economics, Yale University, 2023
B.A., Mathematics and Economics (*with highest distinction*), University of Virginia, 2020

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Ryoichi Sasakawa Young Leaders Fellowship (SYLFF), 2023-2024
Cowles Foundation Fellowship, 2021-2026
Doctoral Fellowship, Yale University, 2021-2026
Best Undergraduate Paper Award, International Atlantic Economic Society, 2020

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Spring 2026, Teaching Assistant to Prof. Peter Schott, Decision Making with Data (MBA), Yale University School of Management
Fall 2025, Teaching Assistant to Prof. Teresa Delgado, Long-Term Fiscal Sustainability (undergraduate), Yale College
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Research Assistant to Prof. Peter Schott and Prof. Teresa Fort, Yale University, 2023-2025
Research Assistant to Prof. Paul Goldsmith-Pinkham, Yale University, Summer 2022

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Data Scientist, 2nd Order Solutions, 2020-2021

**Publications:**

"Selective Entry in Highway Procurement Auctions" (2020), *Atlantic Economic Journal*, 48, 519–533.

**Working Papers:**

"Offshoring and Knowledge Spillovers", *Job Market Paper*

"Workforce Age Structure and Frontier Technology Adoption: Evidence from Newly Released Software" with Tra Nguyen

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**References:**


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## Dissertation Abstract

**Offshoring and Knowledge Spillovers *[Job Market Paper]***

Since the 1990s, U.S. manufacturers have moved production of goods they once made at home to China. Although offshoring lowered their labor costs, their Chinese affiliates and suppliers also gained the know-how required for production. This production knowledge may have diffused to nearby producers, strengthening Chinese companies that export to the United States in competition with U.S. manufacturers. While import competition from China has received substantial attention, less is known about such offshoring-induced knowledge spillovers and what they may have contributed to China's exports to the United States.

This paper studies offshoring as a channel of knowledge diffusion from the U.S. to China and its implications for U.S. welfare. I construct the first measure of U.S. offshoring built on observed domestic production, covering the universe of U.S. manufacturers in 1997. Combining the Census of Manufactures with the Longitudinal Firm Trade Transactions Database, I match U.S. firms' imports over 1997–2017 to their domestic production at the industry level, and classify a firm's imports in an industry and year as *offshored production* if the importing firm also manufactured in that industry in the U.S. in 1997. Using the buyer–supplier links in the import records, I show that, within product and year, offshoring relationships are larger, longer-lasting, and higher in unit value than other trade relationships, consistent with the transfer of production knowledge. They run mostly through independent Chinese suppliers rather than U.S. affiliates.

Does this knowledge reach Chinese exporters that never receive offshoring orders? Spillovers should be largest within a city and industry: production knowledge is more applicable within an industry than across industries, and diffusion is most likely between nearby firms. I infer diffusion from how these exporters' sales to the U.S. responds when an offshoring order first arrives in their city-industry pair, exploiting variation in the timing of the first orders through a staggered difference-in-differences design. Identification relies on treated and not-yet-treated cells following similar outcome trends absent offshoring, conditional on size-specific trends and sector-city-year fixed effects. Exports to the U.S. by never-in-offshoring producers rise by 16% on average two to five years after the first offshoring order.

To evaluate the welfare consequences in each country, I develop a trade model of offshoring and knowledge diffusion, where U.S. firms choose whether to offshore production to China. Offshoring lowers labor costs but also increases China's knowledge stock. Firms do not internalize that the knowledge offshoring passes on to China lowers the prices U.S. consumers pay, while eroding the profits of other U.S. firms due to stronger Chinese competition. As a result, firms may offshore more or less than the welfare-maximizing level for the United States. A per-firm tax on offshoring corrects

this externality without the price distortion of a tariff, while protection of trade secrets removes it without changing offshoring.

**Workforce Age Structure and Frontier Technology Adoption: Evidence from Newly Released Software,** with Tra Nguyen

This paper provides causal evidence that aging slows technology adoption and sheds light on why. Using Lightcast job postings, we track local adoption of 15 software technologies following their releases, instrumenting local age composition with a Bartik-style measure based on historical age structure and national survival rates. Employers in older labor markets post increasingly fewer vacancies requiring the newest software: at year 4, a one-standard-deviation increase in the older-worker share reduces new-technology postings by about 15% of pre-release legacy postings share. The gap is also larger for more difficult technologies, which we measure using new LLM-based difficulty indices. We then develop a model to separate three potential channels through which aging affects technology adoption: shorter work horizon, legacy technology experience loss, and reduced learning efficiency. Removing experience loss reduces the age adoption gap by about 88%, more than removing differences in horizons or learning efficiency. The model also shows that policies affecting workforce age structure have different implications for adoption. Relative to replacement-level fertility with retirement at 70, and assuming zero net migration, a total fertility rate of 1.6 lowers aggregate adoption by 7.5%, whereas raising the retirement age to 75 has almost no effect.